

Reinforcement Learning

Bing-Yi JING
荆炳义

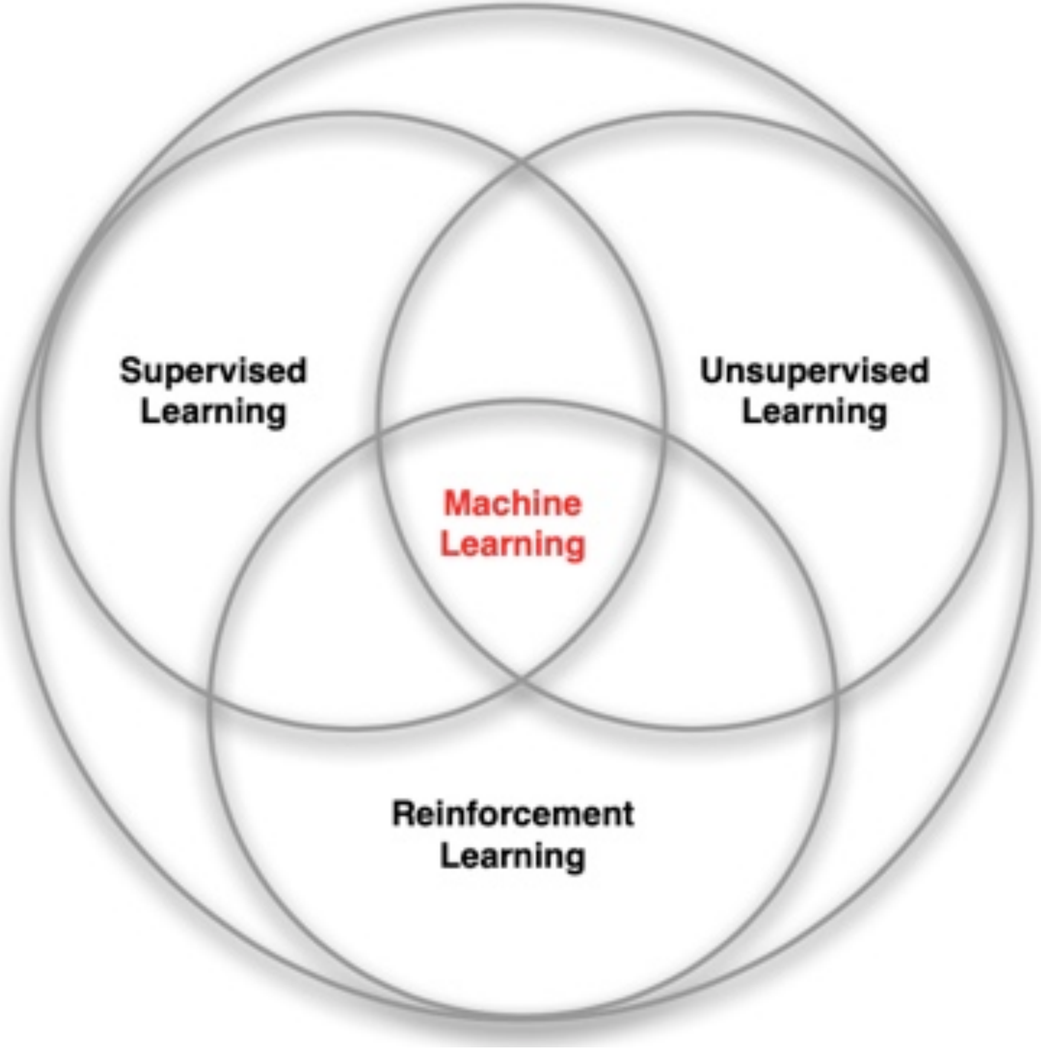
majing@ust.hk

Agenda

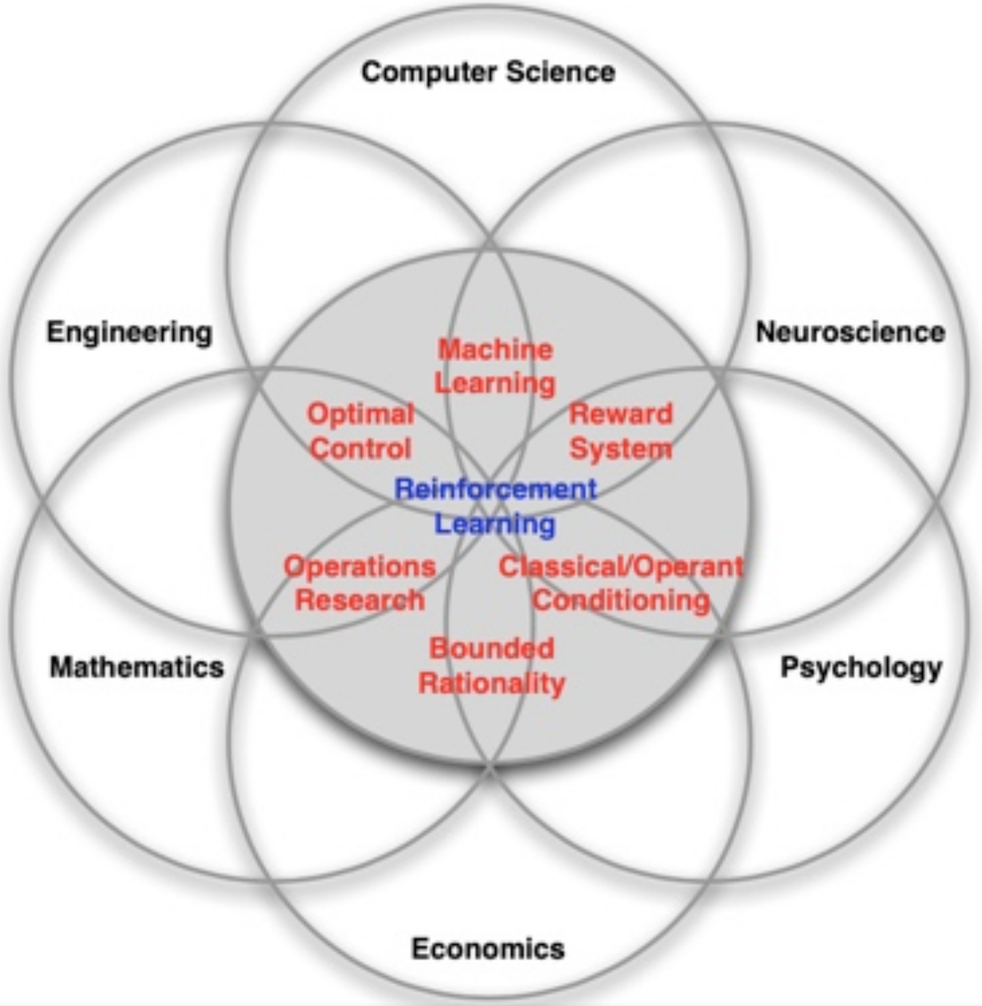
- RL = decision under uncertainty
- Basic concepts

RL = decision under uncertainty

Branches of Machine Learning



RL is ubiquitous



Objective of RL

- Learning to make a “good” sequence of decisions under uncertainty to achieve long-term goal.

Wide applications

- Games
- Recommender system (health, commerce)
- Robotics
- Self-driving
- Smart city (traffic control)
- Career planning
- Business (advertising)

Concrete examples

- Fly stunt manoeuvres in a helicopter
- Backgammon (defeating the world champion)
- Investment portfolio management
- Power station control
- Make a humanoid robot walk
- Atari games (better than humans)





Decision making: heart of our daily life

- We constantly make decisions in uncertain world
- Different decisions lead to different consequences (causal relation)
- Balance between short-term or long-term goal
- Try new things, and learn from mistakes (exploration/exploitation)

Basic Concepts

Characteristics of RL

- Reward
- Optimization
- Delayed feedback
- Exploration/exploitation
- Generalisation

- Policy
- Online / offline learning
- Sequential data (non i.i.d.)
- Sequential decision making

Reward

- A reward R_t is a feedback to a decision (action)
- Indicates how well the agent is doing at step t
- The agent's job is to maximise cumulative reward

Examples

- Backgammon: +/- reward for winning/losing a game
- Atari: +/- reward for increasing/decreasing score
- Portfolio management: + reward for each $ made

Reward Hypothesis

- All goals can be described by the maximisation of expected cumulative reward

Is reward all you need?

Reward is enough

Silver, Singh, Precup, Sutton

Artificial Intelligence, 2021, October

Issues with Reward

- How to define reward functions?
- How to deal with multiple reward functions?
- What if no award is given?
- (Imitation learning: only examples are provided)

Optimization

- Goal: select actions to maximise total future reward
- Need rewards to guide decisions
- e.g. finding shortest route to airport

Delayed feedback

- Decisions may have long-term consequences

Examples

- Go game: don't know if a move is good till the end
- Choosing research topic: easy or difficult one?
- Career planning: work now or further study
- A financial investment (may take months to mature)
- Refuelling a helicopter (to prevent a crash later)
- Blocking opponent moves (might help winning chances many moves from now)

Challenges

- Planning: Balance between short-term vs long-term reward
  - It may be better to sacrifice immediate reward to gain more long-term reward
- Learning: credit assignment is hard (causal relation)
  - Hard to quantify how good/bad a decision is now

探索 从错误中总结

Exploration/Exploitation (trial & error)

- Heart of Reinforcement Learning
- Learning by making decisions and adjustment (trial & error)

How do we proceed?

Exploration	❖ Gather experience by exploring the world
Exploitation	❖ Use experience to guide future decisions

Examples

- Fresh PhD: trying different research problems
- Learning to ride a bike: trying and falling
- Finding good restaurants: trying various places

Examples

- Restaurant Selection
  - Exploitation Go to your favourite restaurant
  - Exploration Try a new restaurant
- Online Banner Advertisements
  - Exploitation Show the most successful advert
  - Exploration Show a different advert
- Oil Drilling
  - Exploitation Drill at the best known location
  - Exploration Drill at a new location
- Game Playing
  - Exploitation Play the move you believe is best
  - Exploration Play an experimental move

Challenges

Censored data

- One trajectory only
- Experience life only once, may have regrets (Counterfactual Regret, CFR)
- e.g., chose HKUST. What'd happen if choosing Tencent?





Inefficiency

- How to explore: Critical
- Huge literature on this
- e.g., value-based, policy-based, Actor-Critic, …

Online vs offline learning

Online

- Learn from own experience (suitable for AI games)

Offline

- Learn from other's experience (everyone is different, needs adjustment)

maybe fixed dataset

Policy

maximize policy

- Mapping from past experience to action

(4) Generalization

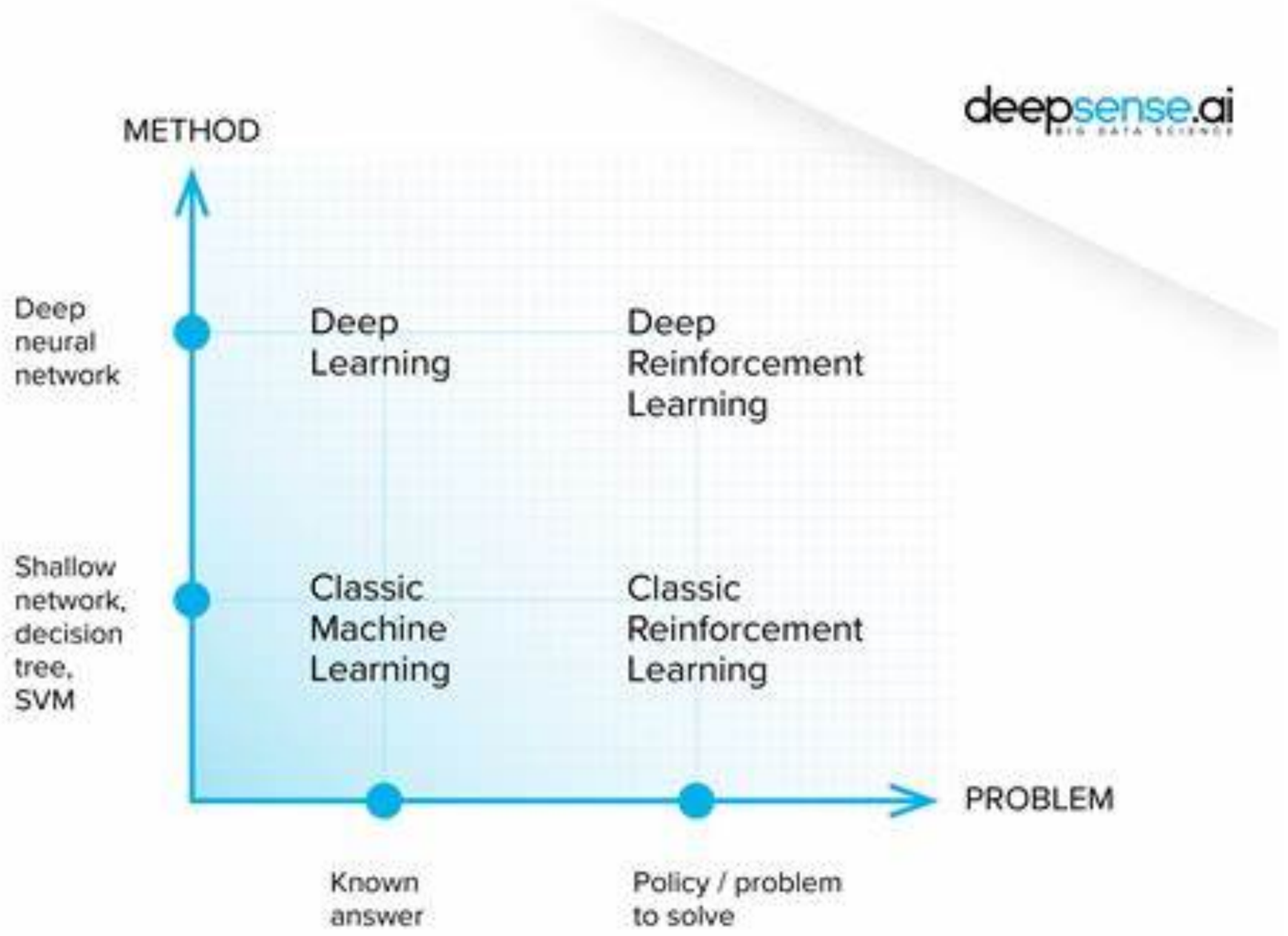
- Small # of states/actions: enumerations/Q-table
- Too many states: generalisation
  - e.g. Machine Learning/Deep Learning

- DeepMind Nature, 2015



- Possible images: [256^(100×200)]^3

DRL = Deep Learning + Reinforcement Learning



RL vs other learnings

one step decision

- SL = Supervised learning
- UL= Unsupervised learning

expoert data no reward how to infer the reward function

- IL = Imitation Learning (Inverse RL)

sequence steps decision

- RL = Reinforcement Learning
- AI Planning = Full Markov Decision Process

Does supervising learning work?

- Yes
  - Rule-based
  - Supervised learning $f : S_t \rightarrow a_t$
- No
  - Results are OK. But not optimal. (Only function approximation)
  - Objective: Optimising reward

- AI Planning = assume to know how decision impact environment

	AI Planning (e.g. Dyn. Prog)	SL	UL	RL	IL
Optimisation	X				
Learns from Experience					
Generalization	X				
Delayed conseqeuences	X				
Exploration					

- SL = correct labels provided

	AI Planning (e.g. Dyn. Prog)	SL	UL	RL	IL
Optimisation	X				
Learns from Experience		X			
Generalization	X	X			
Delayed conseqeuences	X				
Exploration					

- UL = no labels provided (self-supervised learning)

	AI Planning (e.g. Dyn. Prog)	SL	UL	RL	IL
Optimisation	X				
Learns from Experience		X	X		
Generalization	X	X	X		
Delayed conseqeuences	X				
Exploration					

- RL = censored labels provided

	AI Planning (e.g. Dyn. Prog)	SL	UL	RL	IL
Optimisation	X			X	
Learns from Experience		X	X	X	
Generalization	X	X	X	X	
Delayed conseqeuences	X			X	
Exploration				X	

- IL = reduces RL to SL. (Or: Inverse RL + RL)

	AI Planning (e.g. Dyn. Prog)	SL	UL	RL	IL
Optimisation	X			X	X
Learns from Experience		X	X	X	X
Generalization	X	X	X	X	X
Delayed conseqeuences	X			X	X
Exploration				X	

